

Selected Research Papers: Web Analytics, Bot & Crawler Detection

Original curated set of 9 · title + authors + venue match each link · ISWA

Selection (from research session)

These nine papers were chosen from the shared research list: two surveys, log/workload classification, large e-commerce analytics, real-time detection, biostatistics ML, mouse biometrics + logs, a modern multi-stage framework, and TLS fingerprinting. Each entry below has the correct paper title and a matching URL (not a different paper).

1. An Overview of Web Robots Detection Techniques

Hanlin Chen, Hongmei He, Andrew Starr · 2020 · IEEE Cyber Security Conference

Why: Decade survey of web robot/crawler detection: log analysis, ML, hybrids. Hybrid methods outperform single classifiers; online/real-time detection still weak. Best field entry point.

Link: <https://ieeexplore.ieee.org/document/9138856/>

Publisher page (IEEE Xplore; may need access)

2. Web Robot Detection Techniques: Overview and Limitations

Derek Doran, Swapna S. Gokhale · 2011 · Data Mining and Knowledge Discovery (Springer)

Why: Foundational survey. Groups methods into syntactic logs, traffic patterns, ML learning, and Turing tests (CAPTCHA). Explains why UA-only heuristics fail.

Link: <https://link.springer.com/article/10.1007/s10618-010-0180-z>

DOI 10.1007/s10618-010-0180-z · also: ResearchGate / CORE Scholar metadata

3. Classification of Web Robots: An Empirical Study Based on Over One Billion Requests

Junsup Lee, Sungdeok Cha, Dongkun Lee, Hyungkyu Lee · 2009 · Computers & Security

Why: Workload + resource-type metrics on Microsoft.com logs (~1B requests). Separates text crawlers, link checkers, icon crawlers from humans — classic analytics feature paper.

Link: <https://www.sciencedirect.com/science/article/abs/pii/S0167404809000546>

ScienceDirect abstract page (full text may need access)

4. Detecting and Characterizing Web Bot Traffic in a Large E-commerce Marketplace

Xu et al. · 2018 · ESORICS (Springer LNCS)

Why: Taobao/Tmall scale. EM feature selection + gradient trees; characterises bot vs human search/click behaviour for analytics cleaning.

Link: <https://www.eecis.udel.edu/~hnw/paper/esorics18.pdf>

Free full PDF (author-hosted)

5. Efficient On-the-Fly Web Bot Detection

Grazyna Suchacka, Alberto Cabri, Stefano Rovetta, Francesco Masulli · 2021 · Knowledge-Based Systems

Why: Real-time sequential classification of HTTP streams as bot vs human (NN + SPRT). Fits live analytics without waiting for full sessions.

Link: <https://air.unimi.it/retrieve/dfb5ca6f-adb0-468b-93d5-16454650ace2/1-s2.0-S0950705121003373-main.pdf>

Free full PDF (University of Milan OA)

6. New Biostatistics Features for Detecting Web Bot Activity on Web Applications

Rizwan Ur Rahman, Deepak Singh Tomar · 2020 · Computers & Security

Why: Hybrid unsupervised (2-medoid) + supervised ML. Biostatistics features: typing pace, time on page/fields, mouse vs Tab navigation — general bot vs human signals.

Link: <https://www.sciencedirect.com/science/article/abs/pii/S0167404820302741>

ScienceDirect abstract page (full text may need access)

7. Detection of Advanced Web Bots by Combining Web Logs with Mouse Behavioural Biometrics

Iliou et al. · 2021 · ACM Digital Threats: Research and Practice (DTRAP)

Why: Server logs + mouse dynamics against sophisticated browser-mimicking bots. Public dataset for follow-on work.

Link: <https://eprints.bournemouth.ac.uk/36055/1/3447815.pdf>

Free full PDF · ACM DOI 10.1145/3447815

8. BOTracle: A Framework for Discriminating Bots and Humans

Jan Kadel, August See, Ritwik Sinha, Mathias Fischer · 2024 · arXiv:2412.02266 (ESORICS Workshops)

Why: Multi-stage pipeline: heuristics → technical SGAN → behaviour DGCNN graphs. ~40M monthly e-commerce visits; ~98% precision/recall/AUC.

Link: <https://arxiv.org/pdf/2412.02266.pdf>

Free full PDF (arXiv) · abstract <https://arxiv.org/abs/2412.02266>

9. When Handshakes Tell the Truth: Detecting Web Bad Bots via TLS Fingerprints

Jarad & Bicakci · 2026 · arXiv:2602.09606

Why: JA4 TLS fingerprints + gradient boosting (CatBoost/XGBoost). Protocol-level signal harder to spoof than User-Agent.

Link: <https://arxiv.org/pdf/2602.09606.pdf>

Free full PDF (arXiv) · abstract <https://arxiv.org/abs/2602.09606>

Read order: 1 → 2 (surveys) → 3 → 4 (features & e-commerce) → 5 → 6 (online + biostats) → 7 → 8 → 9 (biometrics, BOTracle, TLS).
Free full PDFs: #4, #5, #7, #8, #9. Publisher/abstract pages: #1–3, #6 (institutional access may be required).